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DEPARTMENT OF CSE, CS&IT, AI&DS
COURSE CODE: 24SDCS02 / 24SDCS02E / 24SDCS02P / 24SDCS02L
FULL STACK APPLICATION DEVELOPMENT

Date of the Session: __/__/_____ **Time of The Session: _____to_____**

#SKILL - 5 ➔ Spring Autowiring Demo using @Autowired

Prerequisites:

- **General Idea on Spring Framework**
- **Modules of Spring Framework**
- **Understanding of Autowiring concepts**

A college application maintains **student** information along with their **certification details**. The **Student** object contains a **Certification** object, and instead of creating this Certification object manually, Spring should automatically provide it using the **@Autowired** annotation. When the required classes are marked with **@Component**, Spring creates their objects and manages them inside the **IoC container**. **Using @Autowired**, the Certification object is supplied to the **Student** object by Spring without using the **new keyword**, allowing both objects to be linked through **annotation-based configuration.**

Tasks:

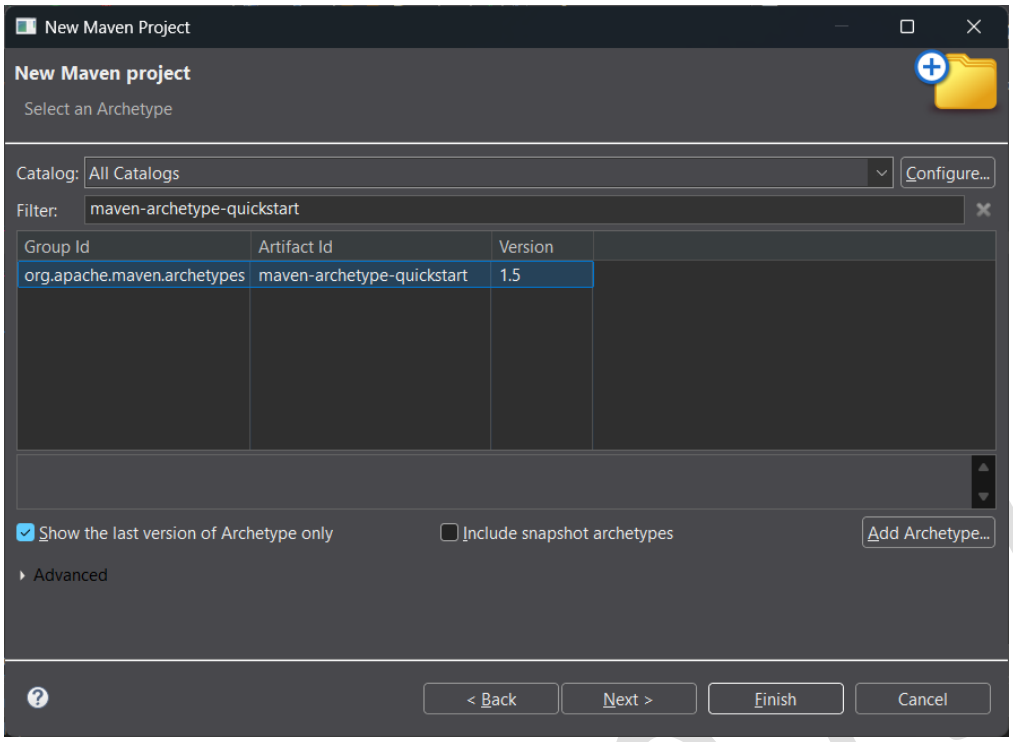
1. Create a **Certification** class with fields such as id, name, and dateOfCompletion.
2. Create a **Student** class with fields such as id, name, gender, and a Certification object.
3. Annotate both classes with **@Component** to enable Spring detection.
4. Use **@Autowired** on a field, constructor, or setter inside the Student class to inject the Certification object.
5. Configure component scanning using either:
   a. An XML configuration file
   b. A Java configuration class with @Configuration and @ComponentScan
6. Load the Spring IoC container using ApplicationContext.
7. Retrieve the Student bean and print all details, including the injected Certification object.
8. **Push the Spring project to GitHub.**

Aim:-

To demonstrate Spring Autowiring using the @Autowired annotation by injecting a Certification object into a Student object using annotation-based configuration.

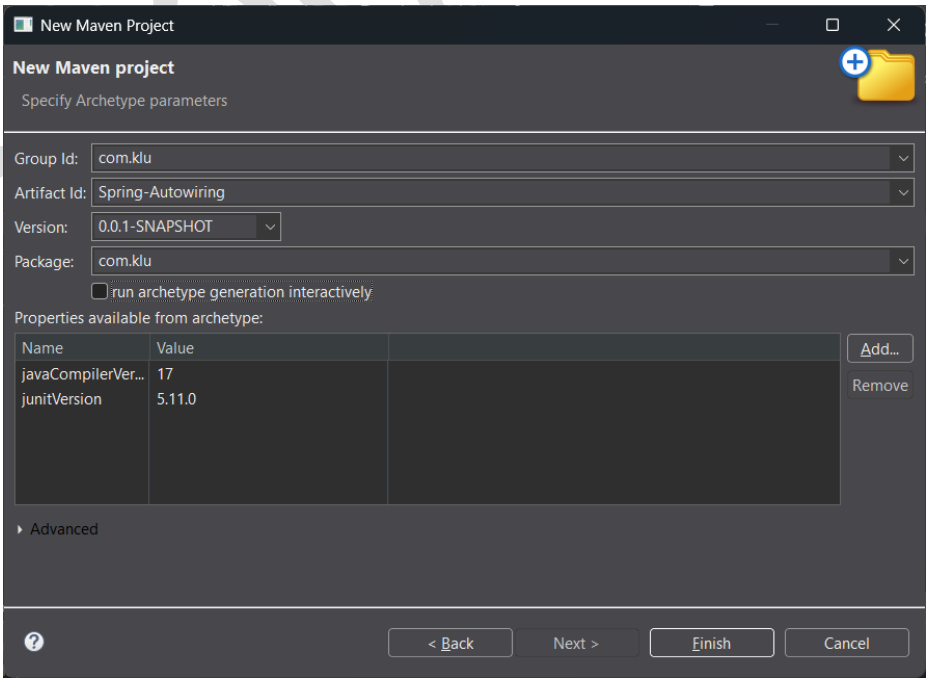
❖ Prerequisites:-

- Basic knowledge of Spring Framework
- Understanding of Spring modules
- Knowledge of Autowiring concepts



Enter details:

- **Group Id**: com.klu
- **Artifact Id**: Spring-Autowiring
- **Packaging**: com.klu
- **Finish**



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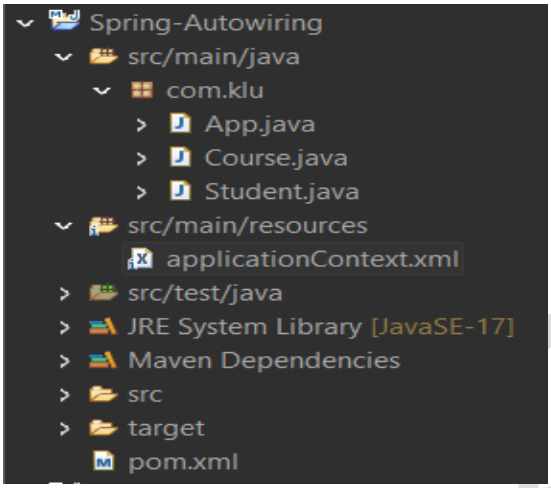
❖ Project Structure:-

```
com.klu
  ├── App.java
  ├── Student.java
  ├── Certification.java
src/main/resources
    ---applicationContext.xml
```



Add Spring Dependency (pom.xml):-

```
<!-- Spring Context -->
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>5.3.30</version>
</dependency>
```

Course.java:-

```
package com.klu;

import org.springframework.stereotype.Component;

@Component
public class Course {

   private int id = 301;
```

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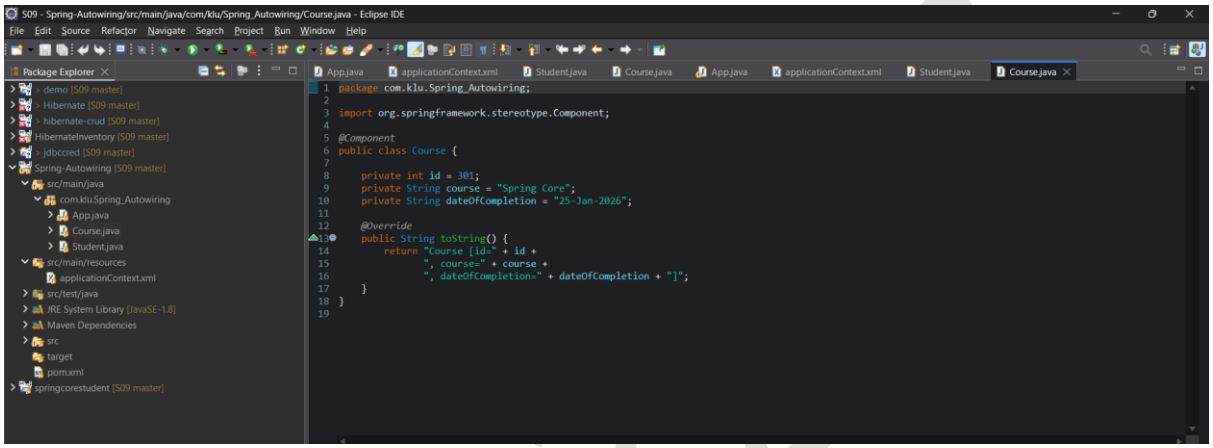
```
    private String course = "Spring Core";
    private String dateOfCompletion = "25-Jan-2026";

    @Override
    public String toString() {
        return "Course [id=" + id +
                ", course=" + course +
                ", dateOfCompletion=" + dateOfCompletion + "]";
    }
}
```



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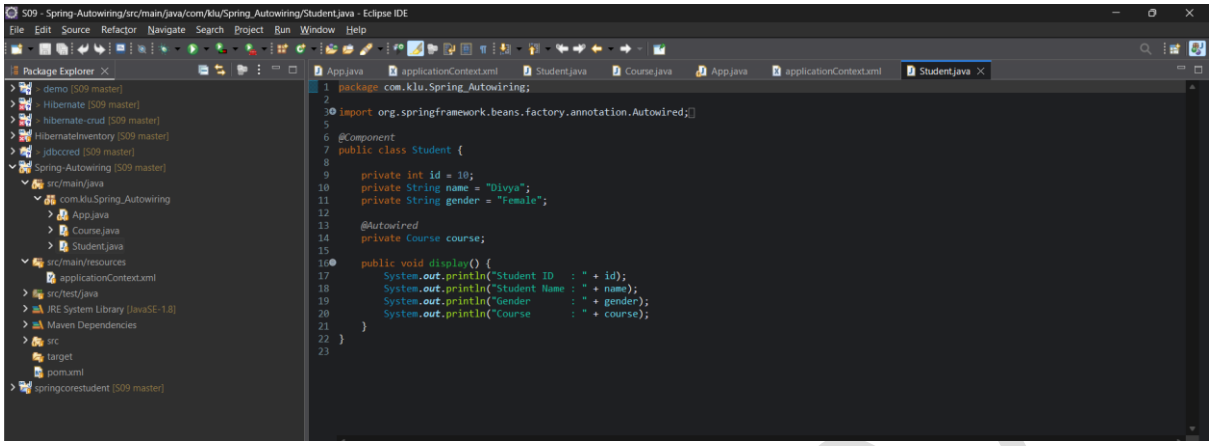
Student.java:-

```
package com.klu;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component
public class Student {

    private int id = 10;
    private String name = "Divya";
    private String gender = "Female";

    @Autowired
    private Course course;

    public void display() {
        System.out.println("Student ID   : " + id);
        System.out.println("Student Name : " + name);
        System.out.println("Gender       : " + gender);
```

```
        System.out.println("Course       : " + course);
    }
}
```



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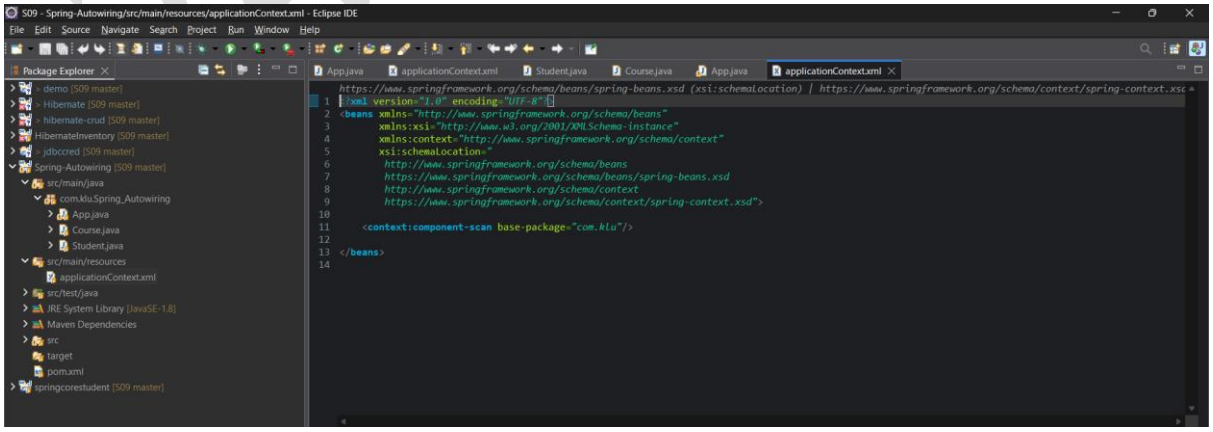
applicationContext.xml

🗀 src/main/resources/applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:context="http://www.springframework.org/schema/context"
    xsi:schemaLocation="
     http://www.springframework.org/schema/beans
     https://www.springframework.org/schema/beans/spring-beans.xsd
     http://www.springframework.org/schema/context
     https://www.springframework.org/schema/context/spring-context.xsd">

  <context:component-scan base-package="com.klu"/>

</beans>
```



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App.java:-

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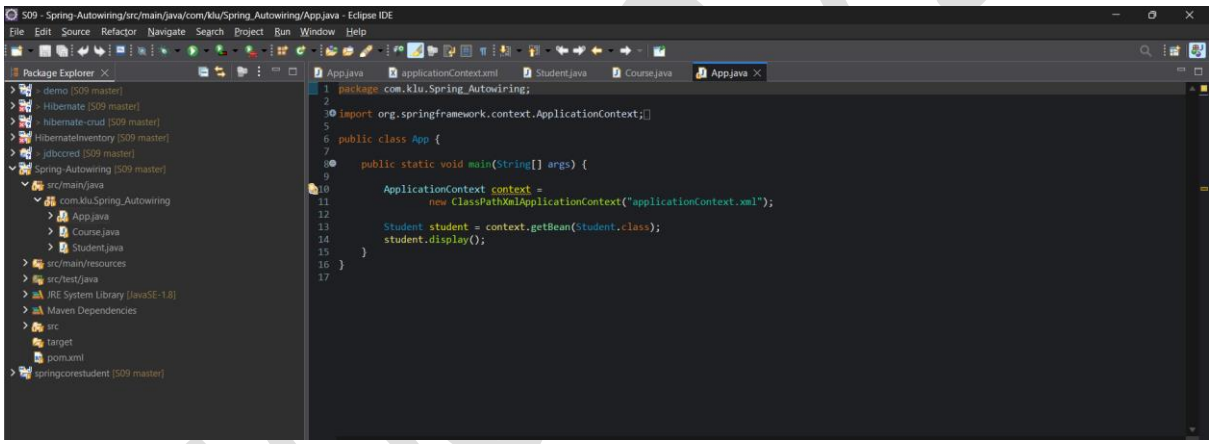
```
package com.klu;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {

    public static void main(String[] args) {

        ApplicationContext context =
                new ClassPathXmlApplicationContext("applicationContext.xml");

        Student student = context.getBean(Student.class);
        student.display();
    }
}
```



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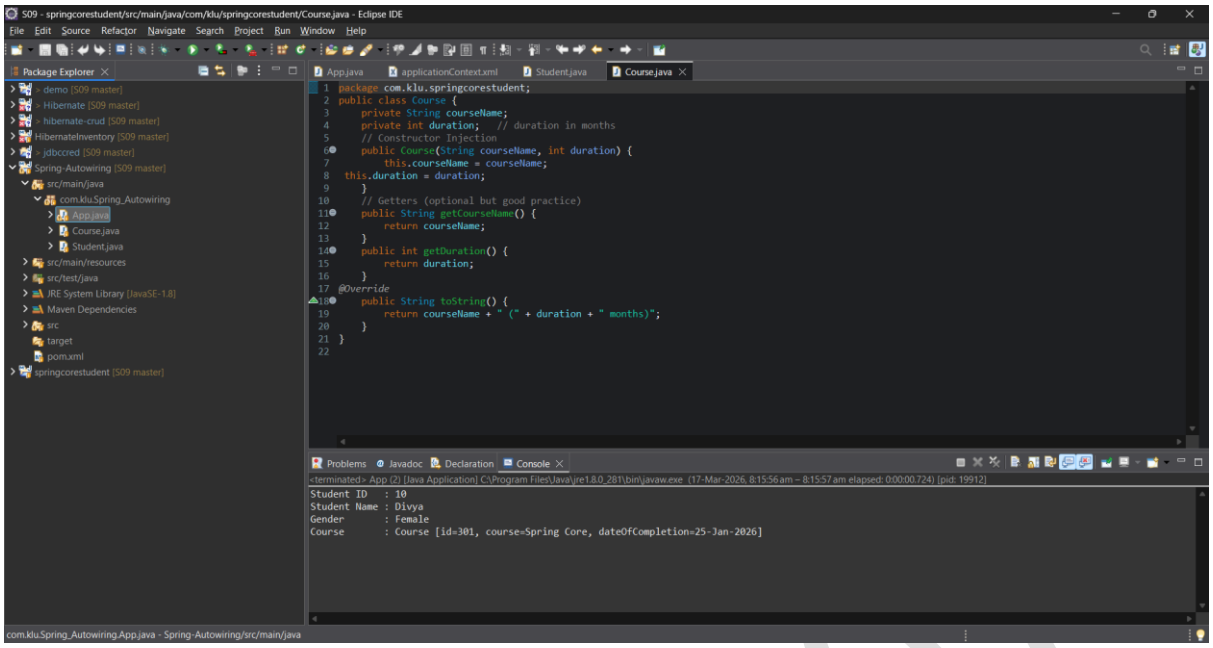
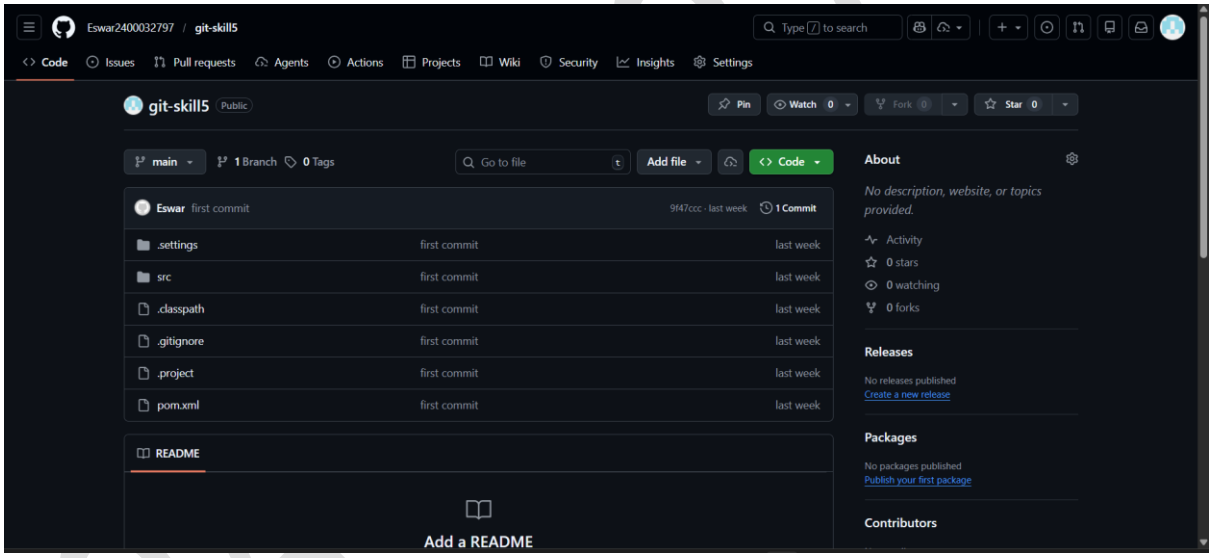
Run the Application:

1. **Right-click App.java**
2. **Run As → Java Application**

OUTPUT on the console:



GIT OUTPUTS:



GIT REPO LINK : https://github.com/Eswar2400032797/git-skill5.git